Avionics. - The mission oriented avionics, by guideline, consist of 1200 pounds of uninstalled weight. This is considered to include communication, identification, navigation, selected computation and selected integrating systems. Other onboard air vehicle avionic equipment such as elements of the flight instruments and flight controls are provided separately in addition to the mission oriented avionics allocation. A 30 percent installation factor is added to the uninstalled avionic weights.

Mass Properties

Composite technology in use at the time of design of 1985 V/STOL short haul transport should provide at least a 25% structural weight saving as prescribed by the study guidelines. Current all composite structural weight savings test data indicate values in excess of 27% are achievable. Use of a blend of metallics for cost reduction, however, with the advanced composites for weight reduction may be somewhat more cost effective. Table 2 indicates the estimated metal aircraft structural weights attainable during the 1980's. Allocation of the percentage of weight savings believed practical for each structural item when using advanced composites is shown along with the resulting estimated advanced composite structure weights. The basic composite

TABLE 2. COMPOSITE WEIGHT SAVINGS BREAKDOWN

	METAL AIRCRAFT WEIGHT	% WT SAVING	1985 ADV. COMPOSITE AIRCRAFT WEIGHT
Structure:			
Wing	6933	25.7	5150
Horizontal	1151	25.7	850
Vertical	935	24.6	705
Fuselage	12543	27.4	9105
Landing Gear	4251	26.4	3130
Surface Controls	2220		2220
Engine Section	4904	27.6	3550
Total	32937	25.0	24715

structural weight technology definition represented by the data of Table 2 was developed from the information presented in Reference 4.

The wing represents a large portion of the structural weight and is subjected to higher than usual torque loads due to the lift fan pods. A more detailed structural analysis was made of the wing structure to validate the estimated weights. Figure 17 presents a diagram of the actual wing structure planform, pertinent design parameters and a breakdown of the weight by structural component. Except for the torque loads introduced by the lift fan pods,